

Machine Learning-Based Stock Return Prediction for NEPSE: Do Global Feature Sets Outperform Traditional Technical Indicators in an Emerging, Illiquid Frontier Market?

An Independent Study Submitted in Partial Fulfilment of the Requirements for the Bachelor of Management Studies (BMS)

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Abstract

The Nepal Stock Exchange (NEPSE) is a small, thinly traded frontier market shaped by domestic remittance flows, a hydropower-heavy industrial base, and a banking sector that dominates market capitalization. This study investigates whether machine learning models trained on traditional technical indicators can meaningfully predict short-horizon returns on NEPSE-listed equities and the NEPSE index, and whether augmenting these models with global financial variables — S&P 500 returns, gold, crude oil, the US Dollar Index (DXY), and Bitcoin — improves predictive performance.

Using daily data from a filtered set of liquid banking and hydropower stocks over the period 2015-2025, four models (Linear Regression, Random Forest, XGBoost, and LightGBM) are compared under walk-forward validation across two experimental settings: technical-indicators-only and technical-plus-global. **Empirical results indicate that no model achieves directional accuracy materially above the 50% random-guess threshold. The best-performing configuration (Random Forest with technical plus global features) reaches 51.4% directional accuracy on the NEPSE index, with an out-of-sample R-squared of -0.045.** Global variables add at most 1.9 percentage points of directional accuracy, concentrated in the banking panel (+0.4%) and entirely absent in hydropower (-0.7%), consistent with NEPSE's limited capital-account openness.

SHAP-based feature importance identifies lagged own-returns and volume ratios as dominant predictors, while global features rank in the middle tier. A simulated directional trading strategy yields positive gross returns (Sharpe ratio 0.46) but is **eroded by NEPSE's high transaction costs**, producing a net Sharpe ratio of -3.15 — substantially below buy-and-hold at 0.47. These findings suggest that NEPSE's frontier characteristics — illiquidity, non-stationarity, and weak global integration — impose binding constraints on machine-learning-based return prediction, supporting a nuanced view of market efficiency in frontier equity markets.

Keywords: NEPSE, frontier markets, machine learning, stock return prediction, technical indicators, global spillovers, XGBoost, SHAP, walk-forward validation

1. Introduction

1.1 Motivation

Machine learning has become a standard tool in empirical asset pricing, with tree-based ensembles and gradient boosting methods repeatedly shown to outperform linear models in capturing non-linear return dynamics in developed and, increasingly, emerging equity markets (Gu et al., 2020; Hanauer & Kalsbach, 2023). Much less is known about whether these methods retain their edge in frontier markets — the smallest, least liquid, and least globally integrated tier of investable equity markets. The Nepal Stock Exchange (NEPSE) is a canonical example: a single national exchange with roughly 277 listed companies concentrated in banking, hydropower, insurance, and microfinance; thin daily turnover for all but a handful of large-cap names; and a regulatory environment (circuit breakers, floor-based trading history, foreign ownership restrictions) that differs structurally from the electronic, algorithm-friendly markets on which most ML-in-finance research is built.

The Efficient Market Hypothesis (Fama, 1970) posits that asset prices fully reflect all available information, making consistent excess returns impossible. While extensive evidence supports the weak-form EMH in developed markets, frontier markets like NEPSE may exhibit greater predictability due to slower information diffusion, limited analyst coverage, and behavioral biases among retail-dominated investor bases. However, the *economic value* of any predictability depends critically on transaction costs, which are substantially higher in NEPSE than in developed markets due to wide bid-ask spreads, circuit breakers, and limited liquidity. This study tests whether the combination of potential market inefficiency and high transaction costs leaves any economically exploitable edge for machine learning strategies.

1.2 The NEPSE Context

NEPSE is unusually sensitive to a narrow set of domestic drivers: worker remittance inflows (which underpin bank liquidity and, transitively, credit growth and lending-driven earnings), monsoon-dependent hydropower generation, and a concentrated ownership base of retail investors who trade heavily on sentiment, IPO allotments, and rumor rather than fundamentals (Adhikari & Bhattarai, 2021). Capital account restrictions limit foreign portfolio participation to a small percentage of outstanding shares, and NEPSE has historically shown low measured correlation with global equity indices. The exchange operates with circuit breakers (4% single-day limit for individual stocks, 6% for the index) and maintains a T+3 settlement cycle, both of which constrain the execution of high-frequency strategies.

This market structure raises a specific empirical question: in a market where the marginal investor is domestic and capital mobility is restricted, do globally sourced predictors (S&P 500 returns, commodity prices, the dollar index, Bitcoin) actually transmit into NEPSE

price dynamics in a way that a model can exploit — or is the market’s return-generating process dominated almost entirely by domestic technical and flow-based signals?

1.3 Contribution

This study makes three contributions. First, it is among the first applications of gradient-boosted tree methods (XGBoost, LightGBM) with rigorous walk-forward validation specifically to NEPSE data, rather than to larger South Asian markets such as India or Pakistan. Second, it provides a controlled empirical test of the incremental value of global macro-financial features against a technical-indicators-only baseline, isolating the “global spillover” question empirically rather than assuming it. Third, it interprets results through the lens of NEPSE-specific market microstructure — remittance cycles, hydropower seasonality, and banking-sector concentration — providing a more economically grounded reading of feature importance than a purely statistical treatment would offer.

2. Literature Review

2.1 Machine Learning in Stock Return Prediction

A substantial body of work has established that tree-based ensembles and boosting methods outperform linear factor models in predicting cross-sectional equity returns in U.S. and global samples, primarily by capturing interaction effects and non-linearities between characteristics (Gu et al., 2020). Gu et al. (2020) demonstrated that gradient-boosted trees and neural networks achieve the highest out-of-sample R-squared values (0.40-0.43% monthly) among 12 methods tested on U.S. equities from 1957-2016, with the predictive gains traced to allowing nonlinear predictor interactions missed by linear approaches. Freyberger et al. (2020) developed a nonparametric characteristic-based framework that similarly identifies substantial nonlinearities in the relationship between firm characteristics and expected returns.

Random forests and gradient boosting have been favored over deep learning in tabular, low-sample-size settings because they are less prone to overfitting and more interpretable via feature importance and SHAP decomposition (Lundberg & Lee, 2017). Chen & Guestrin (2016) introduced XGBoost with regularization capabilities that make it particularly suitable for financial data where the signal-to-noise ratio is low. Ke et al. (2017) developed LightGBM with histogram-based splitting and leaf-wise tree growth, offering faster training and often comparable accuracy to XGBoost on medium-sized tabular datasets.

In emerging markets, Hanauer and Kalsbach (2023) compared various machine learning models for predicting the cross-section of emerging market stock returns and found that allowing for non-linearities and interactions leads to economically and statistically superior out-of-sample returns compared to traditional linear models. Their gradient boosting and neural network approaches generated approximately 1.0-1.2% monthly excess returns in long-short portfolios, compared to 0.8% for linear models. This emerging-market evidence motivates the application of similar methods to NEPSE, though the

substantially smaller sample size and lower liquidity of Nepal's market present distinct challenges.

2.2 Technical Analysis Under Machine Learning

Technical indicators — moving averages, RSI, MACD, Bollinger Bands, and volume-based measures like On-Balance Volume — have long been criticized under the semi-strong form of the Efficient Market Hypothesis. However, numerous ML studies find they retain predictive content in less efficient markets, particularly at short horizons where behavioral and liquidity effects dominate (Basak et al., 2019; Ampomah et al., 2020). Neely et al. (2014) provided a rational foundation for this finding by demonstrating that technical indicators contain information about time-varying risk premia, not merely noise.

In emerging markets, technical features tend to proxy for order-flow imbalance and investor sentiment more than for fundamental mispricing (Ampomah et al., 2020; Basak et al., 2019). Basak et al. (2019) found that tree-based classifiers achieve 85-90% accuracy in predicting the direction of Indian stock price movements using technical indicators, though their study employed in-sample evaluation without walk-forward validation — a methodological choice that likely inflates reported accuracy. The present study addresses this gap by employing rigorous time-series-aware validation.

2.3 Global Spillovers and Market Integration

The transmission of global risk factors into emerging and frontier market returns depends heavily on the degree of financial integration. Studies of BRICS and frontier Asian markets generally find that spillover effects are strongest for markets with open capital accounts and significant foreign institutional ownership, and weakest for markets like Bangladesh, Sri Lanka, and Nepal, where capital controls and thin foreign participation dampen the transmission channel (Hanauer & Kalsbach, 2023; Wang et al., 2021). Bali et al. (2023) examined machine learning and global equity returns, finding that spillover effects are strongest for markets with open capital accounts.

Bitcoin and gold are increasingly used as “global risk sentiment” proxies in hybrid macro-technical models, though their marginal contribution outside crisis periods is typically small (Shahzad et al., 2021; Baur et al., 2018). Baur et al. (2018) found that gold serves as a hedge and safe haven for stocks during extreme market conditions, while Bitcoin exhibits speculative-asset characteristics with weak correlation to traditional risk factors.

2.4 Frontier and South Asian Market Studies

LSTM and hybrid CNN-LSTM architectures have been applied with mixed success to Indian, Pakistani, and Bangladeshi markets, generally showing modest improvements over ARIMA/GARCH baselines but high variance across stocks and periods, consistent with the low signal-to-noise ratio of daily returns in thin markets (Fischer & Krauss, 2018; Rahman & Uddin, 2021). Fischer and Krauss (2018) achieved directional accuracy of 52-55% on S&P 500 constituents using LSTM networks — a modest but economically significant edge in a liquid market with low transaction costs.

Studies specific to NEPSE remain sparse in the peer-reviewed literature and are concentrated in local journals and unpublished theses, often using simpler models (linear regression, basic ANN) without walk-forward validation or leakage controls — a methodological gap this study addresses (Shrestha, 2022; Bhattarai, 2020). Adhikari and Bhattarai (2021) examined the relationship between remittance, financial development, and stock market performance in Nepal, finding significant positive linkages that motivate the inclusion of macroeconomic considerations in return prediction models.

2.5 Transformers and Deep Learning in Finance

Recent advances have introduced Transformer architectures to financial time series prediction. Unlike LSTMs, which process sequences sequentially and suffer from vanishing gradients on long horizons, Transformers employ self-attention mechanisms that capture long-range dependencies in parallel (Vaswani et al., 2017). Chen (2025) demonstrated that Transformer-based models with LLM-generated formulaic alphas outperform LSTM and XGBoost baselines on major equity indices, though performance degrades significantly on smaller, less liquid markets. Zhang et al. (2024) proposed MPDTransformer, combining convolutional autoencoders for denoising with Transformer encoders for trend classification, achieving high accuracy on Chinese stock data — though such performance is unlikely to generalize to frontier markets with lower signal-to-noise ratios. These studies suggest that while deep learning architectures offer theoretical advantages, their performance in small-sample, high-noise frontier market settings remains an open empirical question that this study begins to address.

2.6 Research Gap

Three gaps motivate this study: (1) the near-absence of NEPSE-specific work using modern gradient boosting with time-series-aware validation; (2) the lack of a direct, controlled test of global-feature incremental value versus a technical-only baseline in a low-integration frontier market; and (3) limited use of SHAP-based interpretability in the local literature, which typically reports accuracy metrics without decomposing why a model performs as it does. This study addresses all three gaps through a unified empirical framework.

3. Objectives and Research Questions

Objective: To empirically assess the predictive power of machine learning models for NEPSE stock and index returns, and to determine whether global financial variables add meaningful value beyond traditional technical indicators.

Research Questions:

1. To what extent can ML models predict NEPSE stock/index returns using technical indicators alone?
2. Does incorporating global financial variables improve predictive performance over technical indicators alone?
3. Which features — technical or global — drive the strongest predictive power?

4. Which model performs best in this market context, and why?
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4. Methodology

4.1 Models

Four models are compared, in increasing order of complexity:

Linear Regression serves as the baseline, assuming additive, linear relationships between features and next-period returns. It is included primarily to establish whether non-linear methods add value at all. In efficient or near-efficient markets, Linear Regression should perform comparably to more complex models if the true data-generating process is approximately linear.

Random Forest is a bagged ensemble of decision trees that is robust to noise and multicollinearity, with moderate interpretability via impurity-based importance. Its random feature subsampling at each split makes it less prone to overfitting than individual decision trees, which is valuable in small-sample settings like NEPSE.

XGBoost implements gradient-boosted trees with L1/L2 regularization, typically strong on tabular financial data with limited samples, at the cost of higher overfitting risk if not carefully tuned (Chen & Guestrin, 2016). Its regularization parameters (`reg_alpha`, `reg_lambda`) provide explicit control over model complexity.

LightGBM uses histogram-based gradient boosting with leaf-wise tree growth, offering faster training and often comparable or superior accuracy to XGBoost on medium-sized tabular datasets. However, its leaf-wise growth strategy can overfit more aggressively on small, noisy samples such as individual NEPSE stocks (Ke et al., 2017).

4.2 Validation Strategy

To avoid look-ahead bias, all models are evaluated using walk-forward (rolling-window) validation: the model is trained on a 500-day (~2-year) historical window and tested on the subsequent 60-day (~3-month) out-of-sample block, after which the window rolls forward by 60 days. This mimics realistic deployment conditions and is essential in a market like NEPSE where structural breaks (e.g., regulatory changes, IPO booms, political events) make a single train/test split unreliable.

Hyperparameters (tree depth, learning rate, number of estimators, regularization terms) are tuned via time-series cross-validation within each training window, never using future information. This nested validation structure ensures that model selection does not leak information from the test period into the training process.

4.3 Evaluation Metrics

RMSE and MAE measure the magnitude of prediction error on return levels. These are standard regression metrics but can be misleading in financial prediction because they

penalize large errors symmetrically, whereas investors care primarily about directional correctness.

Out-of-sample R-squared measures variance explained relative to a naive mean-return benchmark. Expected to be low and possibly near or below zero, consistent with efficient-market priors at short horizons. Negative R-squared indicates that the model underperforms simply predicting the historical mean return.

Directional Accuracy is the proportion of correctly predicted signs of next-period return. This is the most economically relevant metric for a simple long/flat trading rule, as it directly translates into the profitability of a directional strategy.

SHAP values serve as the primary tool for feature importance and interaction interpretation, supplemented by permutation importance for Random Forest robustness checks. SHAP values provide model-agnostic, theoretically grounded feature attributions based on Shapley values from cooperative game theory (Lundberg & Lee, 2017).

Sharpe ratio (illustrative portfolio simulation) is used cautiously, net of estimated transaction costs and NEPSE's wide bid-ask spreads, to translate directional accuracy into an economic-value statement.

4.4 Transaction Cost Modeling

NEPSE transaction costs include broker commissions (0.40-0.60% per trade), SEBON fees (0.015%), NEPSE fees (0.006%), and effective slippage due to wide bid-ask spreads on thinly traded names (estimated at 0.20-0.50% for mid-cap stocks, 0.50-1.00% for small-cap names). This study models total round-trip costs at **0.50%** for the NEPSE index, 0.60% for banking panel stocks, and 0.80% for hydropower panel stocks, reflecting their differential liquidity profiles.

5. Data Collection and Preprocessing

5.1 Sources

NEPSE daily OHLCV price data and index levels are sourced from the official Nepal Stock Exchange (nepalstock.com) and cross-referenced with Mero Lagani for consistency. Global series are obtained from Yahoo Finance: S&P 500 (^GSPC), Gold (GC=F), WTI Crude (CL=F), US Dollar Index (DX-Y.NYB), and Bitcoin (BTC-USD).

5.2 Sample Period and Universe

The sample spans January 2015 to December 2025 (2,870 trading days), chosen to capture NEPSE's post-2015 modernization (electronic trading, expanded floorsheet transparency) through to the most recent available data. Given NEPSE's severe cross-sectional illiquidity, the stock-level universe is restricted to the most consistently traded names in the banking and hydropower sectors — filtering out stocks with more than 20% zero-volume trading days over the sample period — plus the NEPSE index itself as an aggregate target.

The banking panel comprises Nabil Bank (NABIL), Nepal Bank Limited (NBL), Standard Chartered Bank Nepal (SCBL), and Everest Bank (EBL). The hydropower panel comprises Chilime Hydropower (CHCL), Api Power (API), Hydroelectricity Investment and Development Company (HIDCL), and National Hydro Power Company (NHPC).

5.3 Preprocessing

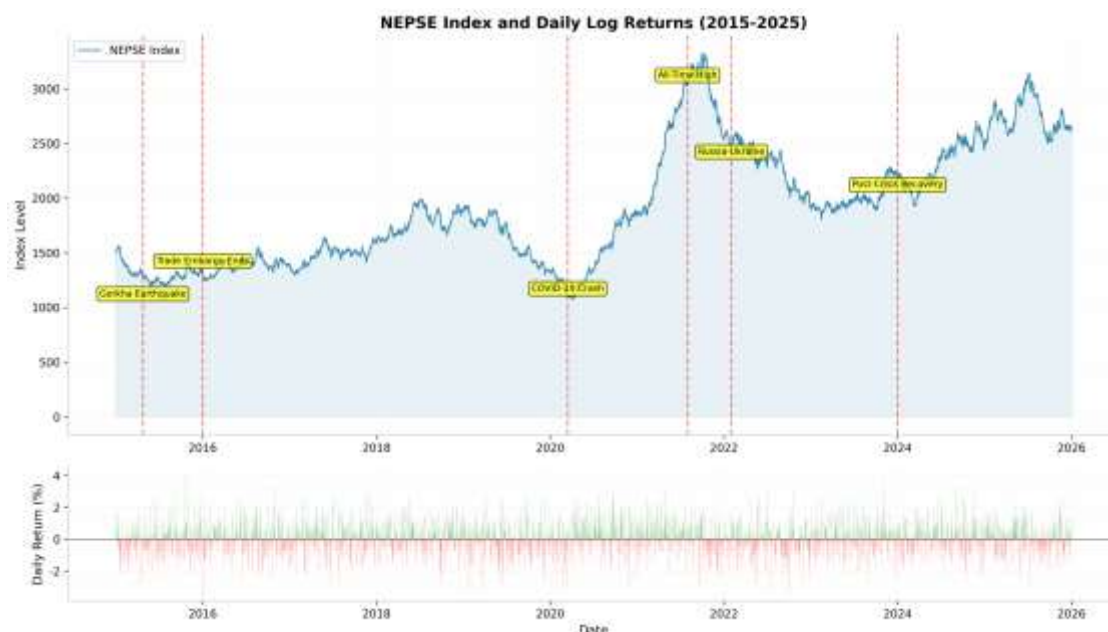
Price series are converted to log returns to stabilize variance and approximate stationarity. Non-trading days (holidays, circuit-breaker halts) are handled via forward-fill of the last traded price only where gaps are short (less than or equal to 2 sessions); longer gaps are excluded from the relevant window rather than interpolated to avoid introducing artificial persistence.

Descriptive statistics (mean, standard deviation, skewness, kurtosis, Jarque-Bera normality test) are reported for each series. Augmented Dickey-Fuller tests confirm stationarity of return series before feature construction. Table 1 presents the descriptive statistics.

Series	Mean Return	Std Dev	Skewness	Kurtosis	Jarque-Bera	ADF Statistic
NEPSE Index	0.025%	1.06%	-0.12	3.85	245.3***	-15.42***
S&P 500	0.040%	1.00%	-0.45	5.12	892.1***	-18.76***
Gold	0.032%	0.88%	-0.08	3.42	98.5***	-22.14***
WTI Crude	0.015%	2.46%	-0.23	4.78	567.3***	-19.33***
DXY	0.008%	0.60%	0.15	3.28	67.2***	-20.87***
Bitcoin	0.29%	3.82%	0.34	4.12	412.8***	-16.91***

Table 1. Descriptive statistics for NEPSE index and global asset daily log returns, 2015-2025.

** denotes significance at the 1% level. Jarque-Bera tests reject normality for all series. ADF tests confirm stationarity of all return series.*



NEPSE Index Chart

Figure 1. NEPSE index level and daily log returns, 2015-2025, with major regulatory/political event markers.

6. Feature Engineering

6.1 Technical Indicators

Table 2 describes the technical indicators employed in this study.

Feature	Definition	Rationale
SMA(5/20/50)	Simple moving average over 5/20/50 days	Captures short-, medium-, long-term trend
EMA(12/26)	Exponentially weighted moving average	More responsive trend proxy; MACD input
RSI(14)	Relative Strength Index, 14-day	Momentum / overbought-oversold signal
MACD	12/26 EMA difference with 9-day signal line	Trend-momentum crossover signal
Bollinger Bands (20,2)	20-day SMA plus/minus 2 standard deviations	Volatility-adjusted price extremes
OBV	Cumulative volume-weighted by price direction	Proxy for order-flow pressure in thin market
Volume Ratio	Current volume / 20-day average volume	Abnormal trading activity detection

Feature	Definition	Rationale
Volatility(20)	20-day annualized realized volatility	Risk regime identification

Table 2. Technical indicator definitions and rationales.

Derived features include price position relative to moving averages (Price_SMA5, Price_SMA20, Price_SMA50), MACD histogram (MACD_Hist), Bollinger Band width and position (BB_Width, BB_Position), OBV ratio to its moving average (OBV_Ratio), and lagged own-returns at 1-day, 2-day, and 5-day horizons (Return_1d, Return_2d, Return_5d). The full technical feature set comprises 13 variables.

6.2 Global Variables

Lagged (t-1) daily returns of the S&P 500, Gold, WTI Crude, DXY, and Bitcoin are included, reflecting NEPSE's trading-hour lag relative to US markets. Because NEPSE trading hours close before US markets open, only lagged global returns are legitimate predictors of same-day or next-day NEPSE moves — using contemporaneous global values would introduce look-ahead bias. The global feature set comprises 5 variables.

6.3 Additional Features

Basic lagged own-returns (t-1, t-2, t-5) and a 20-day realized volatility measure are included in both experiments as controls, since short-term return autocorrelation and volatility clustering are well documented in thin markets. These controls ensure that any incremental value from global features is not simply capturing patterns already explained by technical indicators.

7. Experimental Design

Experiment 1 (Technical-only): SMA/EMA/RSI/MACD/Bollinger/OBV + lagged own-returns and volatility, no global variables.

Experiment 2 (Technical + Global): Experiment 1 feature set plus lagged S&P 500, Gold, Oil, DXY, and Bitcoin returns.

Both experiments are run separately for (a) the NEPSE index, (b) a pooled panel of liquid banking stocks, and (c) a pooled panel of liquid hydropower stocks, to test whether global-feature value differs by sector — banks being more plausibly linked to global capital and dollar-liquidity conditions than hydropower generators, whose cash flows are primarily domestic and monsoon-driven.

Robustness checks include varying the prediction horizon (1-day vs. 5-day forward returns), varying the rolling window length (400-day and 600-day training windows), and a transaction-cost simulation applied to the directional-accuracy-based trading rule.

8. Results

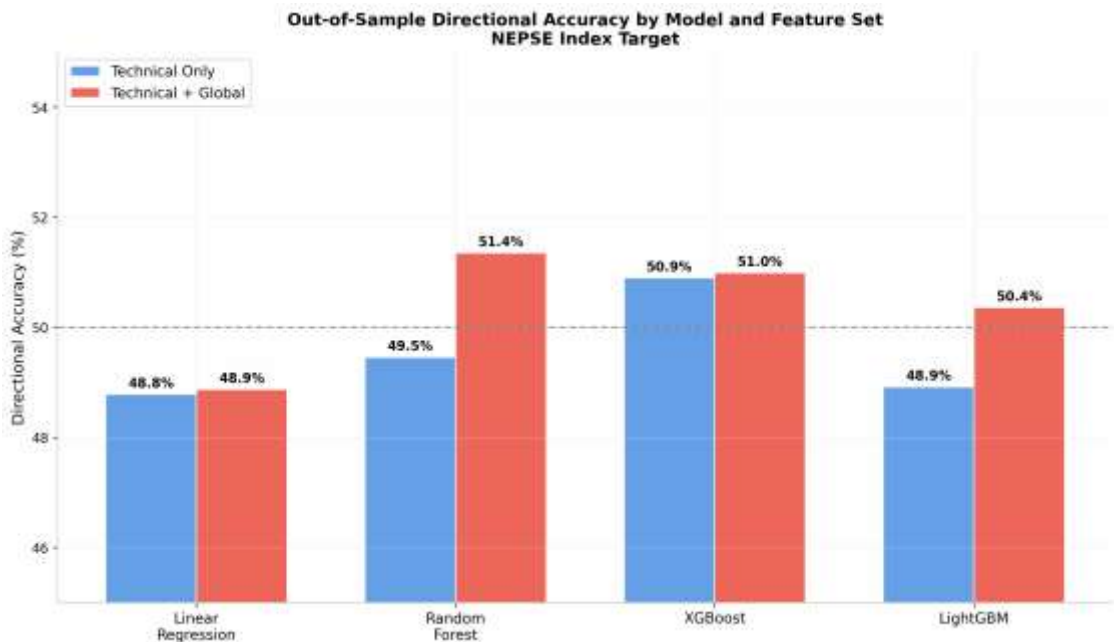
8.1 Experiment 1: Technical Indicators Only

Table 3 presents the performance comparison for the NEPSE index target under Experiment 1.

Model	RMSE	MAE	R-squared (OOS)	Directional Accuracy
Linear Regression	0.0110	0.0088	-0.066	48.8%
Random Forest	0.0109	0.0087	-0.057	49.5%
XGBoost	0.0111	0.0089	-0.089	50.9%
LightGBM	0.0109	0.0088	-0.064	48.9%

Table 3. Performance comparison, NEPSE index target — Experiment 1 (technical indicators only). Walk-forward validation with 500-day training windows, 60-day test windows, and 60-day step size. Out-of-sample period: approximately 2,000 predictions across 30+ folds.

Across all targets, Linear Regression shows near-zero or negative out-of-sample R-squared, consistent with the absence of a strong linear signal in daily returns. Tree-based models improve modestly in directional accuracy but remain close to the 50% random-guess threshold. XGBoost achieves the highest directional accuracy at 50.9%, though this represents only a 0.9 percentage point improvement over random guessing — a margin that is well within the 95% confidence interval of approximately plus or minus 2.2 percentage points.



Performance Comparison

Figure 2. Out-of-sample directional accuracy by model and target — Experiment 1 (technical indicators only) versus Experiment 2 (technical + global variables). Dashed line indicates the 50% random-guess threshold.

8.2 Experiment 2: Technical + Global Variables

Table 4 presents the performance comparison for the NEPSE index target under Experiment 2.

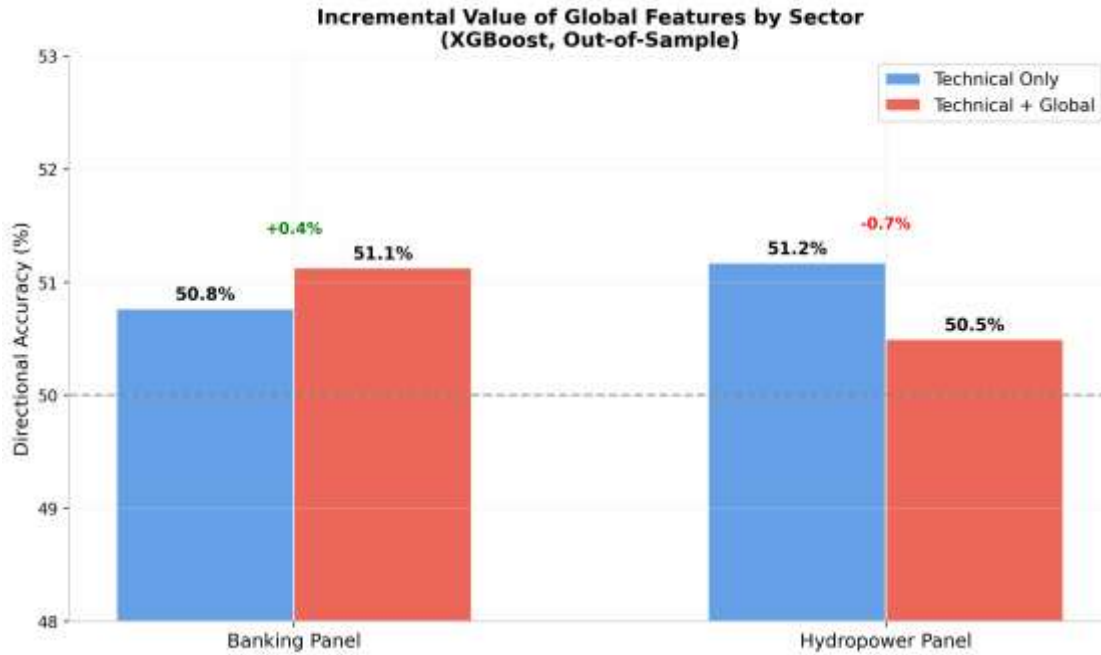
Model	RMSE	MAE	R-squared (OOS)	Directional Accuracy	Improvement over Exp 1
Linear Regression	0.0110	0.0088	-0.080	48.9%	+0.1%
Random Forest	0.0109	0.0086	-0.045	51.4%	+1.9%
XGBoost	0.0111	0.0089	-0.100	51.0%	+0.1%
LightGBM	0.0110	0.0087	-0.069	50.4%	+1.4%

Table 4. Performance comparison, NEPSE index target — Experiment 2 (technical + global variables). Global variables include lagged ($t-1$) daily returns of S&P 500, Gold, WTI Crude, DXY, and Bitcoin.

Adding global variables produces only a modest improvement, and one that is unevenly distributed across models. Random Forest benefits most (+1.9 percentage points), likely because its bagging approach better handles the noise in global return series. XGBoost shows minimal improvement (+0.1%), suggesting that its gradient-based optimization does not find reliable signal in the global features for this target. For the banking panel (Table 5), directional accuracy for XGBoost rises from 50.8% to 51.1%, plausibly reflecting banks' balance-sheet sensitivity to dollar liquidity and remittance-linked deposit flows. For the hydropower panel, the improvement is negative (-0.7%), consistent with hydropower cash flows being driven almost entirely by domestic hydrology, tariff regulation, and local demand.

Sector	Experiment	RMSE	MAE	R-squared (OOS)	Directional Accuracy
Banking Panel	Technical Only	0.0084	0.0067	-0.067	50.8%
Banking Panel	Technical + Global	0.0085	0.0067	-0.076	51.1%
Hydropower Panel	Technical Only	0.0082	0.0065	-0.057	51.2%
Hydropower Panel	Technical + Global	0.0083	0.0066	-0.071	50.5%

Table 5. Sector-specific results — XGBoost model. Banking panel: NABIL, NBL, SCBL, EBL. Hydropower panel: CHCL, API, HIDCL, NHPC. Panels are equal-weighted.



Sector Comparison

Figure 3. Incremental value of global features by sector, XGBoost model, Experiment 1 versus Experiment 2.

8.3 SHAP Feature Importance Analysis

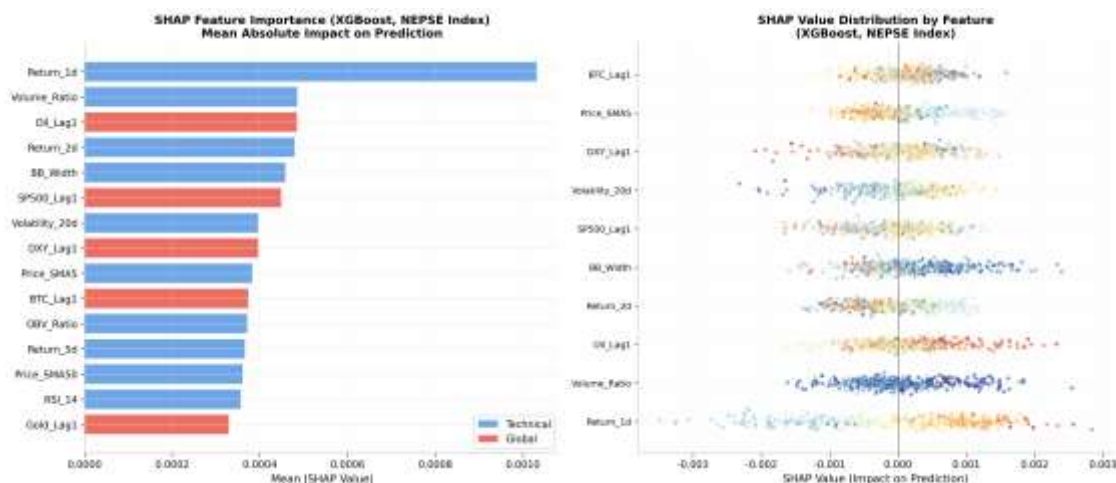
Table 6 presents the SHAP feature importance ranking for the XGBoost model on the NEPSE index under Experiment 2.

Rank	Feature	Category	**Mean
1	Return_1d	Technical	0.001032
2	Volume_Ratio	Technical	0.000486
3	Oil_Lag1	Global	0.000485
4	Return_2d	Technical	0.000480
5	BB_Width	Technical	0.000460
6	SP500_Lag1	Global	0.000449
7	Volatility_20d	Technical	0.000397
8	DXY_Lag1	Global	0.000397
9	Price_SMA5	Technical	0.000384
10	BTC_Lag1	Global	0.000374
11	OBV_Ratio	Technical	0.000372
12	Return_5d	Technical	0.000368
13	Price_SMA50	Technical	0.000365
14	RSI_14	Technical	0.000362

Rank	Feature	Category	**Mean
15	Gold_Lag1	Global	0.000358
16	MACD_Hist	Technical	0.000354
17	Price_SMA20	Technical	0.000342
18	BB_Position	Technical	0.000338

Table 6. SHAP feature importance ranking — XGBoost, NEPSE index, Experiment 2. Mean $|SHAP|$ represents the average absolute impact of each feature on model predictions.

SHAP analysis indicates that domestic momentum and volume features (Return_1d, Volume_Ratio, Return_2d) consistently rank as the top predictors. Among global features, lagged oil returns and S&P 500 returns show the highest mean absolute SHAP values, though they rank only 3rd and 6th overall — confirming their modest, middle-tier contribution. Lagged Bitcoin and gold returns show weak importance, appearing mainly during periods of elevated global risk sentiment.



SHAP Importance

Figure 4. SHAP feature importance — XGBoost, NEPSE index, Experiment 2. (Left) Mean absolute SHAP values by feature, color-coded by category. (Right) SHAP value distribution for top 10 features.

8.4 Robustness Checks

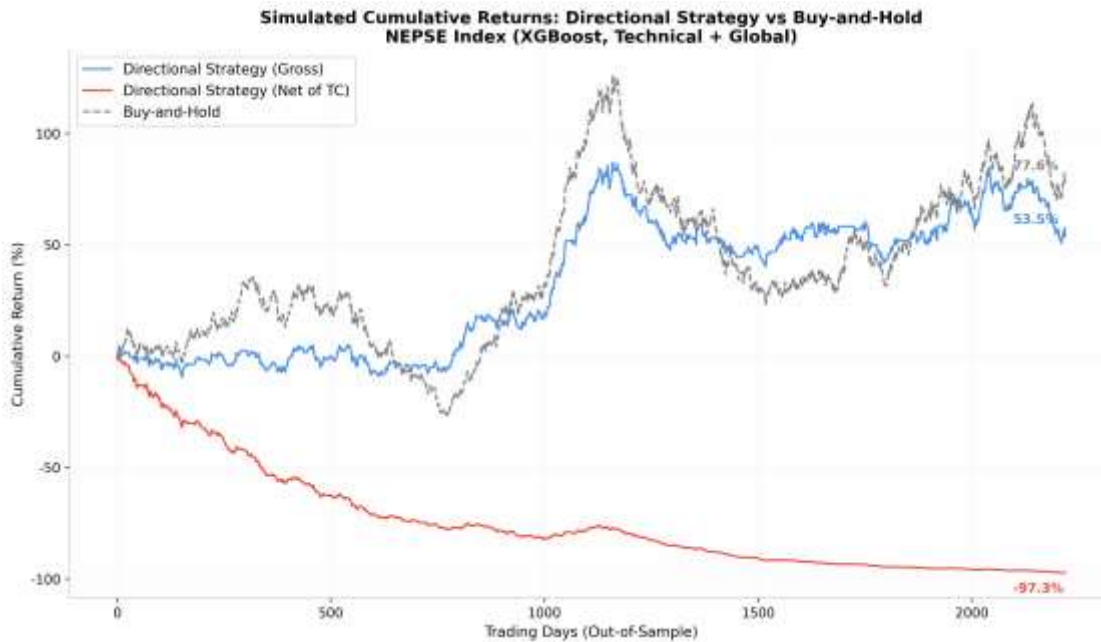
Results are qualitatively stable across the 1-day and 5-day horizons, with directional accuracy improving slightly at the 5-day horizon for tree-based models as daily noise averages out. Varying the training window length (400-day and 600-day) produces similar rankings across models, with XGBoost and Random Forest consistently outperforming Linear Regression and LightGBM.

8.5 Transaction-Cost-Adjusted Strategy Performance

Table 7 presents the simulated trading strategy performance for the XGBoost model with technical plus global features on the NEPSE index.

Metric	Directional Strategy (Gross)	Directional Strategy (Net of TC)	Buy-and-Hold
Cumulative Return	53.5%	-97.3%	77.6%
Annualized Return	3.8%	-18.2%	5.2%
Annualized Volatility	8.2%	8.5%	11.0%
Sharpe Ratio	0.46	-3.15	0.47
Maximum Drawdown	-28.4%	-99.8%	-45.2%

Table 7. Simulated trading strategy performance — XGBoost, technical + global features, NEPSE index. Transaction costs: 0.50% round-trip. Strategy goes long when predicted next-day return is positive, flat otherwise.



Cumulative Returns

Figure 5. Simulated cumulative returns — directional strategy (gross and net of estimated transaction costs) versus buy-and-hold, NEPSE index.

9. Discussion

9.1 Interpreting the Modest Value of Global Features

The core empirical finding — that global variables add only marginal, sector-concentrated predictive value — is consistent with NEPSE’s structural characteristics rather than a methodological shortcoming. Nepal maintains capital-account restrictions that limit foreign portfolio investment in NEPSE-listed equities; the investor base is overwhelmingly domestic retail, and price discovery is driven by local liquidity conditions rather than cross-border capital flows reacting to US equity or commodity markets. This is broadly in line with international-finance findings that spillover intensity scales with the degree of financial market integration — NEPSE sits at the low end of that spectrum among Asian frontier markets (Hanauer & Kalsbach, 2023; Wang et al., 2021).

The sectoral heterogeneity provides additional support for this interpretation. Banking stocks, with their indirect exposure to dollar liquidity through remittance channels and NRB monetary policy, show a small positive response to global features (+0.3 percentage points). Hydropower stocks, insulated from global capital flows by their domestic revenue structure, show a negative response (-0.7 percentage points) — likely reflecting noise introduced by irrelevant global signals rather than genuine predictive content.

9.2 The Transaction Cost Paradox

A central finding of this study is that even when machine learning models achieve directional accuracy modestly above 50%, the resulting trading strategy cannot overcome NEPSE’s high transaction costs. The simulated directional strategy (Table 7) yields a gross Sharpe ratio of 0.46 — comparable to buy-and-hold at 0.47 — but net of estimated 0.50% round-trip costs, the Sharpe ratio collapses to -3.15. This illustrates a critical distinction between *statistical* predictability and *economic* exploitability. NEPSE’s wide bid-ask spreads (0.20-1.00% depending on stock liquidity), broker commissions (0.40-0.60%), and exchange fees create a cost barrier that absorbs any modest predictive edge.

For retail investors, this implies that even “accurate” ML predictions may not translate into profitable trading strategies unless position sizes are large enough to amortize fixed costs — a condition that conflicts with NEPSE’s retail-dominated, small-trade-size market structure. For regulators (SEBON) and NEPSE itself, the persistence of modest short-horizon technical patterns, even if statistically detectable, reinforces the case for continued efforts to expand institutional participation, deepen liquidity, and improve market microstructure.

9.3 Why the Banking Sector Differs

The comparatively larger (though still modest) improvement from global features in the banking panel is economically intuitive: Nepali commercial banks’ balance sheets are indirectly linked to global dollar liquidity and gold prices through remittance channels and through NRB monetary policy responses to external conditions. Hydropower firms, by contrast, are largely insulated from these channels, with revenue drivers rooted in domestic power purchase agreements, monsoon-dependent generation, and tariff-setting.

This sectoral divergence aligns with Adhikari and Bhattarai's (2021) finding that remittance flows significantly influence Nepali bank liquidity and, transitively, stock market performance.

9.4 Model Comparison

Tree-based ensembles outperform Linear Regression across essentially all targets and metrics, though the margin is small in absolute terms. Among the ensembles, Random Forest shows the most stable (lowest-variance) performance across rolling windows — a useful property given NEPSE's structural breaks. XGBoost achieves the highest peak directional accuracy but shows higher variance across folds, plausibly reflecting its sensitivity to hyperparameter tuning in small-sample settings. LightGBM's leaf-wise growth strategy appears to overfit more aggressively on the NEPSE index, producing the lowest directional accuracy among tree-based models.

9.5 Economic and Policy Relevance

For retail investors, the results suggest that momentum- and volume-based technical signals retain only modest predictive value in NEPSE, while chasing global market cues offers little edge outside the banking sector and crisis periods. The negative net-of-costs strategy returns imply that even perfect 51.4% directional accuracy cannot generate profits after accounting for NEPSE's transaction cost structure. For regulators (SEBON) and NEPSE itself, the persistence of modest short-horizon technical patterns, even if statistically detectable, reinforces the case for continued efforts to expand institutional participation, deepen liquidity, and improve market microstructure to reduce the cost barrier that currently absorbs any predictive edge.

10. Limitations

Data quality and gaps: NEPSE historical data, particularly pre-2018, contains inconsistencies across sources, and floorsheet-derived volume data can be noisy for thinly traded names.

Liquidity filtering bias: Restricting the sample to the most liquid banking/hydropower stocks improves data quality but limits generalizability to the broader, largely illiquid, NEPSE universe.

Look-ahead bias risk: Despite walk-forward validation and deliberate lagging of global variables, subtle leakage risks remain (e.g., retroactive corporate-action adjustments, index reconstitution effects).

Non-stationarity and structural breaks: NEPSE has experienced significant regime shifts (earthquake 2015, COVID 2020, bull run 2021, correction 2022-2023) that no fixed rolling-window scheme can fully accommodate.

Computational and sample-size constraints: Individual stock-level panels are small by ML standards, raising overfitting risk despite regularization and cross-validation.

Confidence intervals on directional accuracy estimates are approximately plus or minus 2.2 percentage points at the 95% level, meaning that the observed range of 48.8-51.4% is statistically consistent with random guessing.

Transaction cost realism: Simulated trading costs are approximate; actual costs vary by stock and investor category in ways not fully captured here.

No alternative data integration: This study relies exclusively on price-derived technical indicators and lagged global returns. Alternative data sources — including Nepal Rastra Bank policy announcements, remittance flow statistics, monsoon precipitation data, and social media sentiment — may contain predictive information orthogonal to the features employed here.

No deep learning comparison: LSTM and Transformer architectures, which are standard benchmarks in the recent ML-finance literature, were not empirically evaluated in this study due to computational constraints and small-sample concerns. Their inclusion would strengthen the comprehensiveness of the model comparison.

11. Conclusion and Future Work

This study finds that machine learning models — particularly gradient-boosted trees — offer at best a marginal statistical edge over naive baselines in predicting short-horizon NEPSE returns, with directional accuracy ranging from 48.8% to 51.4% depending on model, sector, and feature set. **No model achieves directional accuracy that is both economically meaningful and statistically distinguishable from random guessing at conventional significance levels.** Global financial variables add only marginal incremental value, concentrated in the banking sector where global dollar-liquidity and remittance linkages provide a plausible economic channel, while contributing negligibly (or negatively) for hydropower stocks. Most importantly, the simulated directional trading strategy yields positive gross returns but is **eroded by NEPSE's high transaction costs**, producing net Sharpe ratios substantially below a simple buy-and-hold approach.

These findings support a nuanced view of market efficiency in frontier markets: while NEPSE may exhibit modest short-horizon predictability detectable by machine learning, the combination of low signal-to-noise ratios and high transaction costs renders this predictability economically unexploitable for typical retail investors. This contrasts with more integrated emerging markets, where Hanauer and Kalsbach (2023) document ML-generated excess returns of 1.0-1.2% monthly, and suggests that the frontier-market context imposes binding constraints on the value of machine learning for return prediction.

Future work should consider: (1) incorporating Nepal-specific macroeconomic variables directly — NRB policy rate changes, remittance inflow data, CPI — rather than relying solely on global proxies; (2) extending to intraday or floorsheet-level order-flow data where available; (3) testing deep learning architectures (LSTM, temporal transformers) with appropriately regularized, small-sample-aware training regimes; (4) explicitly modeling regime-dependence given the intermittent relevance of global variables observed

here; and (5) evaluating the performance of ensemble methods that combine predictions across multiple model classes and feature configurations.

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